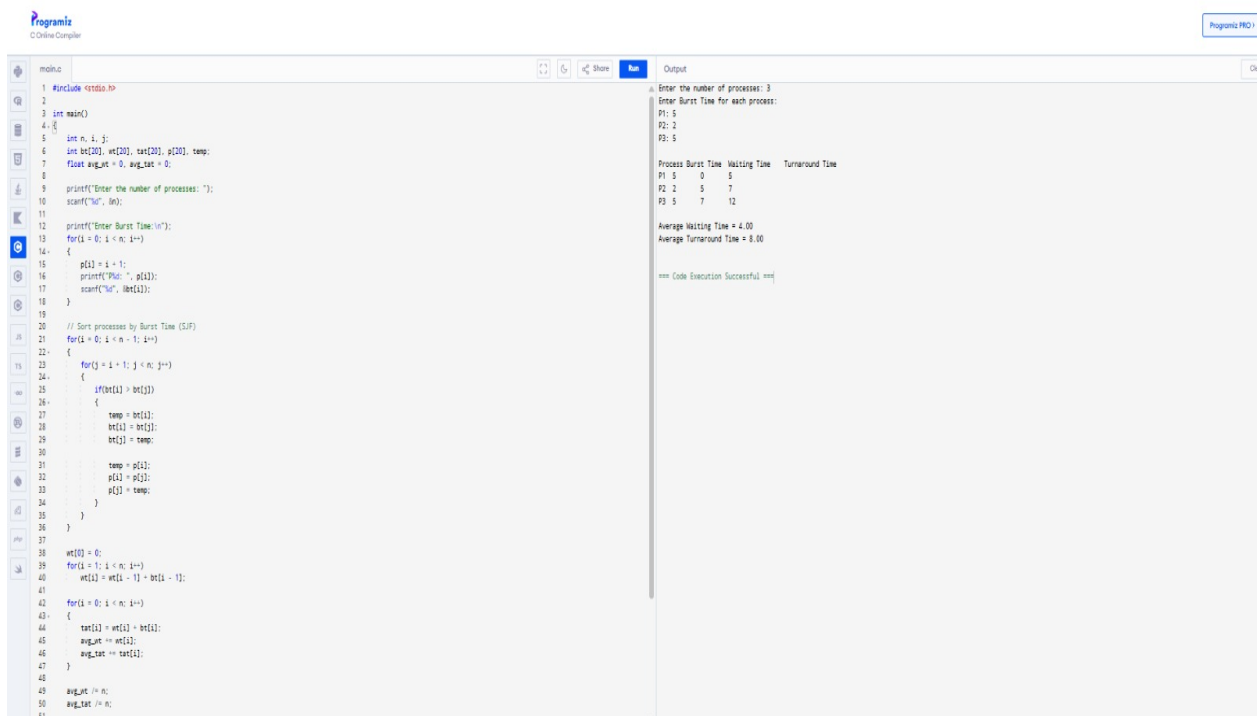


QUESTION 4:

Construct a scheduling program with C that s

CODING:



The screenshot shows a C program in an online compiler. The program implements the Shortest Burst Time (SJT) scheduling algorithm. It takes the number of processes and their burst times as input, sorts them by burst time, and then calculates the waiting and turnaround times for each process. The output shows the results for three processes: P1, P2, and P3.

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int n, i, j;
6     int bt[20], wt[20], tat[20], p[20], temp;
7     float avg_wt = 0, avg_tat = 0;
8
9     printf("Enter the number of processes: ");
10    scanf("%d", &n);
11
12    printf("Enter Burst Time:\n");
13    for(i = 0; i < n; i++)
14    {
15        p[i] = i + 1;
16        printf("P%d: ", p[i]);
17        scanf("%d", &bt[i]);
18    }
19
20    // Sort processes by Burst Time (SJF)
21    for(i = 0; i < n - 1; i++)
22    {
23        for(j = i + 1; j < n; j++)
24        {
25            if(bt[i] > bt[j])
26            {
27                temp = bt[i];
28                bt[i] = bt[j];
29                bt[j] = temp;
30
31                temp = p[i];
32                p[i] = p[j];
33                p[j] = temp;
34            }
35        }
36    }
37
38    wt[0] = 0;
39    for(i = 1; i < n; i++)
40        wt[i] = wt[i - 1] + bt[i - 1];
41
42    for(i = 0; i < n; i++)
43    {
44        tat[i] = wt[i] + bt[i];
45        avg_wt += wt[i];
46        avg_tat += tat[i];
47    }
48
49    avg_wt /= n;
50    avg_tat /= n;
51}
```

Output:

```
Enter the number of processes: 3
Enter Burst Time for each process:
P1: 5
P2: 2
P3: 5

Process Burst Time Waiting Time Turnaround Time
P1 5 0 5
P2 2 5 7
P3 5 7 12

Average Waiting Time = 4.00
Average Turnaround Time = 8.00

=== Code Execution Successful ===
```